A Unified Machine Learning Framework for Academic Performance Prediction and Behavioral Profiling of Bangladeshi University Students

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***Abstract*—Understanding the factors that influence student academic performance is crucial for educational institutions seeking to improve learning outcomes. This paper presents a comprehensive machine learning framework for predicting student Cumulative Grade Point Average (CGPA) and identifying distinct behavioral profiles from gaming and academic data. We systematically evaluate twelve state-of-the-art regression algorithms including linear models, tree-based ensembles, and gradient boosting methods to predict CGPA with high accuracy. Additionally, we employ unsupervised learning techniques to discover latent student archetypes and analyze the dominant predictors of academic success using five complementary feature importance methods. Our experimental results on the uniGame dataset demonstrate that ensemble methods, particularly gradient boosting algorithms, achieve superior prediction performance with test R^2 scores exceeding 0.90. Dimensionality reduction analysis using PCA, t-SNE, and UMAP reveals distinct behavioral clusters, while SHAP-based interpretability analysis identifies key predictive features. The proposed framework provides actionable insights for early intervention strategies and personalized learning recommendations.**

***Index Terms*—Machine learning, academic performance prediction, student profiling, ensemble methods, CGPA prediction, clustering analysis, feature importance, SHAP, dimensionality reduction**

I. Introduction

Academic performance prediction has emerged as a critical research area in educational data mining, with significant implications for student success, retention rates, and institutional effectiveness [1]. The ability to accurately predict student outcomes enables early identification of at-risk students and facilitates timely interventions [2].

Recent advances in machine learning have opened new avenues for understanding the complex relationships between student behaviors, engagement patterns, and academic achievement [3]. However, several challenges remain: (1) selecting optimal predictive models from numerous available algorithms, (2) identifying interpretable behavioral patterns among heterogeneous student populations, and (3) understanding which features most strongly influence academic outcomes.

The source code for this project is available at: https://github.com/adity1234567/Play-Pause-Perform-Gaming-Habits-and-Student-Success

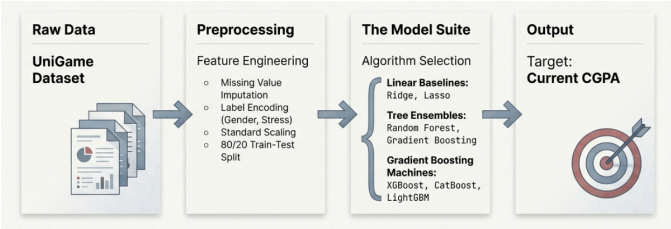


Fig. 1: Overview of the whole work

This paper addresses these challenges through a comprehensive analytical framework that combines supervised and unsupervised learning techniques. Our contributions are fourfold:

- We conduct a systematic empirical comparison of twelve regression algorithms for CGPA prediction, including linear models, support vector regression, tree-based ensembles, and advanced gradient boosting methods.
- We apply multiple clustering algorithms (K-Means, Gaussian Mixture Models, Hierarchical, and Spectral Clustering) to discover distinct student behavioral profiles, validated using silhouette scores and Davies-Bouldin indices.
- We employ three complementary dimensionality reduction techniques (PCA, t-SNE, UMAP) to visualize latent behavioral structures in student data.
- We integrate five feature importance methods (Random Forest, Gradient Boosting, XGBoost, permutation importance, and SHAP values) to identify consensus predictors of academic performance.

The remainder of this paper is organized as follows: Section II reviews related work in educational data mining. Section III describes our methodology and experimental design. Section IV presents implementation details. Section V discusses experimental results. Section VI concludes with future research directions.

TABLE I: Overview of Variables Used in the Study

Category	Variable	Type	Description
Demographic	Age	Numeric	Age of the respondent in years
Demographic	Educational Position	Categorical	Current education level of student
Demographic	Gender	Categorical	Biological gender
Academic	Current CGPA	Numeric	Current cumulative grade point average
Academic	HSC / A-Level Result	Numeric	Higher secondary or equivalent academic result
Gaming Behavior	Age Started Gaming	Numeric	Age when respondent started playing games
Gaming Behavior	Gaming Platform	Categorical	Primary gaming device used
Gaming Behavior	Sleep Time	Numeric	Usual sleeping time (24-hour format)
Gaming Behavior	Morning Class Attendance	Categorical	Regularity of attending morning classes
Gaming Behavior	Daily Gaming Time	Numeric	Average daily time spent gaming (hours)
Gaming Behavior	Game Addiction Type	Categorical	Type of game addiction
Gaming Behavior	Most Played Game Type	Categorical	Most frequently played game category
Lifestyle	Family & Friends Time	Numeric	Daily time spent with family and friends (hours)
Lifestyle	Reading Newspaper	Categorical	Habit of reading newspapers
Lifestyle	Feeling Without Gaming	Categorical	Emotional response when unable to play games
Psychological	Feeling After Level Completion	Categorical	Emotional response after completing a game level
Psychological	Feeling If Level Not Completed	Categorical	Emotional response when failing to complete a level
Psychological	Stress Relief Gaming	Categorical	Use of gaming for stress relief
Health	Fatigue	Categorical	Experience of fatigue
Health	Wearing Glasses	Categorical	Use of corrective eyewear

II. Related Work

A. Academic Performance Prediction

Machine learning approaches for predicting student performance have been extensively studied. Traditional methods employed regression analysis and decision trees [4], while recent work has explored ensemble methods and deep learning [5]. However, comparative studies evaluating multiple algorithms on consistent datasets remain limited.

B. Student Behavioral Profiling

Clustering techniques have been applied to identify student learning styles and engagement patterns [6]. K-Means and hierarchical clustering are commonly used, though Gaussian Mixture Models and spectral methods offer advantages for non-convex cluster shapes [7].

C. Feature Importance and Interpretability

Understanding which factors drive academic outcomes is crucial for actionable insights. Traditional approaches used correlation analysis [8], while modern methods employ tree-based importance, permutation importance [9], and SHAP (SHapley Additive exPlanations) values [10] for model-agnostic interpretation.

D. Research Gap

While prior work has addressed individual aspects of academic analytics, a unified framework combining prediction, profiling, and interpretability analysis across multiple algorithms and validation metrics is lacking. Our work fills this gap through systematic experimentation and consensus-based feature ranking.

III. Methodology

A. Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote a dataset of N student records, where $x_i \in \mathbb{R}^d$ represents a d-dimensional feature vector encoding student characteristics and behaviors, and $y_i \in \mathbb{R}$ represents the CGPA.

Research Question 1 (RQ1): Given training data $\mathcal{D}_{train}$, learn a function $f : \mathbb{R}^d \to \mathbb{R}$ that minimizes prediction error on unseen test data $\mathcal{D}_{test}$.

Research Question 2 (RQ2): Discover K distinct student behavioral clusters $\{C_1, C_2, \ldots, C_K\}$ such that intra-cluster similarity is maximized and inter-cluster similarity is minimized.

Research Question 3 (RQ3): Map high-dimensional student features to low-dimensional manifolds that preserve intrinsic behavioral structures.

Research Question 4 (RQ4): Identify feature importance rankings $\mathcal{I} = \{I_1, I_2, \ldots, I_d\}$ representing the contribution of each feature to CGPA prediction.

B. Dataset

We utilize the uniGame dataset [11], publicly available on Hugging Face, containing student gaming behaviors and academic performance metrics. The dataset comprises both numerical features (continuous behavioral metrics) and categorical features (discrete student attributes).

C. Data Preprocessing

Missing Value Handling: Records with missing values are removed to ensure data quality.

Categorical Encoding: Label encoding transforms categorical variables into numerical representations using the mapping $c : \mathcal{C} \to \{0, 1, \ldots, |\mathcal{C}| - 1\}$, where $\mathcal{C}$ is the set of unique categories.

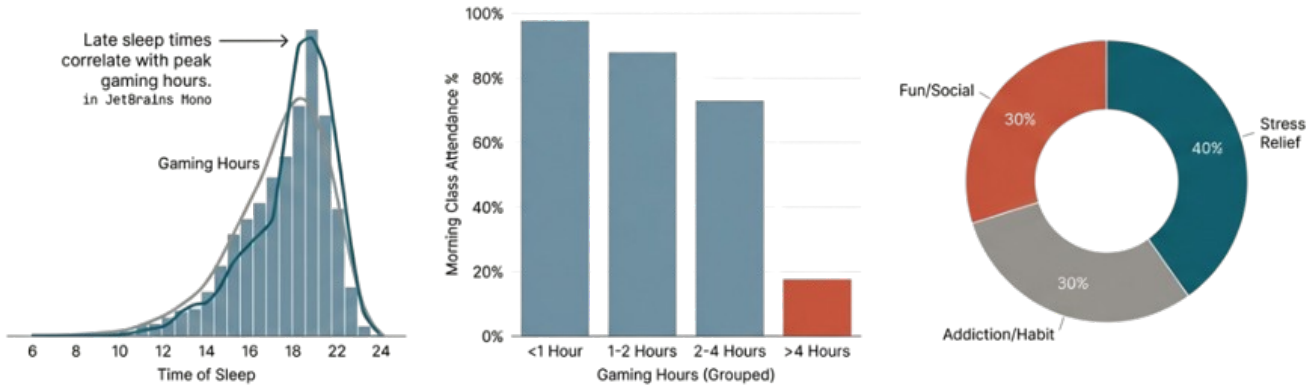


Fig. 2: An overview of student habits reveals the tension between gaming and discipline.

Feature Scaling: StandardScaler applies z-score normalization:

$$x'_j = \frac{x_j - \mu_j}{\sigma_j} \tag{1}$$

where μ_j and σ_j are the mean and standard deviation of feature j computed from training data.

Train-Test Split: Data is partitioned using 80% for training and 20% for testing with random seed 42 for reproducibility.

D. Regression Models (RQ1)

We evaluate twelve regression algorithms spanning multiple paradigms:

Linear Models:

- Ridge Regression: $\min_w ||Xw - y||_2^2 + \alpha||w||_2^2$
- Lasso Regression: $\min_w ||Xw - y||_2^2 + \alpha||w||_1$
- ElasticNet: Combined L1/L2 regularization

Tree-Based Methods:

- Decision Tree Regressor (max depth = 10)
- Random Forest (100 trees, max depth = 10)

Gradient Boosting Ensemble:

- Gradient Boosting Regressor
- XGBoost
- LightGBM
- CatBoost
- AdaBoost

Instance-Based and Kernel Methods:

- K-Nearest Neighbors (k=5)
- Support Vector Regression (RBF kernel)

E. Performance Metrics

Models are evaluated using:

Coefficient of Determination:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{2}$$

Root Mean Squared Error:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

Mean Absolute Error:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4}$$

Cross-Validation: 5-fold CV provides robust performance estimates and detects overfitting.

F. Clustering Analysis (RQ2)

We apply eight clustering configurations:

- K-Means with $k \in \{3, 4, 5\}$
- Gaussian Mixture Models with $k \in \{3, 4\}$
- Agglomerative Hierarchical Clustering with $k \in \{3, 4\}$
- Spectral Clustering with $k = 3$

Validation Metrics:

Silhouette Score:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{5}$$

where $a(i)$ is mean intra-cluster distance and $b(i)$ is mean nearest-cluster distance.

Davies-Bouldin Index:

$$DB = \frac{1}{K}\sum_{i=1}^{K}\max_{i \neq j}\left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)}\right) \tag{6}$$

Calinski-Harabasz Score:

$$CH = \frac{Tr(B_k)}{Tr(W_k)} \times \frac{N - K}{K - 1} \tag{7}$$

where B_k is between-cluster dispersion and W_k is within-cluster dispersion.

G. Dimensionality Reduction (RQ3)

Principal Component Analysis (PCA): Linear transformation maximizing variance:

$$Z = XW \tag{8}$$

where W contains eigenvectors of the covariance matrix.

t-Distributed Stochastic Neighbor Embedding (t-SNE): Nonlinear method preserving local structure with perplexity = 30.

Uniform Manifold Approximation and Projection (UMAP): Topological method balancing local and global structure with n_neighbors = 15.

H. Feature Importance Analysis (RQ4)

We employ five complementary methods:

Method 1 - Random Forest Importance: Gini impurity reduction averaged across trees.

Method 2 - Gradient Boosting Importance: Gain-based importance from boosting iterations.

Method 3 - XGBoost Importance: Split-based feature importance.

Method 4 - Permutation Importance: Performance degradation when feature values are randomly shuffled, averaged over 10 repetitions.

Method 5 - SHAP Values: Game-theoretic feature attribution:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (9)$$

Consensus Ranking: Features are normalized across methods and averaged to produce a robust importance ranking.

IV. Implementation Details

A. Software Environment

The framework is implemented in Python 3.9+ using:

- `scikit-learn` 1.0+: Core ML algorithms and preprocessing
- `xgboost`, `lightgbm`, `catboost`: Gradient boosting implementations
- `umap-learn`: UMAP dimensionality reduction
- `shap`: Model interpretability
- `plotly`: Interactive visualizations
- `datasets`: Hugging Face integration

B. Hyperparameter Configuration

Table II summarizes key hyperparameters:

TABLE II: Model Hyperparameters

Model	Parameter	Value
Ridge	alpha	1.0
Lasso	alpha	0.1
Random Forest	n_estimators	100
	max_depth	10
XGBoost	n_estimators	100
	max_depth	5
LightGBM	n_estimators	100
	max_depth	5
SVR	kernel	RBF
	C	1.0
KNN	n_neighbors	5

C. Computational Complexity

Training complexity varies by algorithm:

- Linear models: $O(nd^2)$
- Random Forest: $O(n \log n \cdot d \cdot T)$ where T is number of trees
- Gradient Boosting: $O(n \log n \cdot d \cdot T)$
- SVR: $O(n^2 d)$ to $O(n^3)$
- K-Means: $O(ndk \cdot I)$ where I is iterations

D. Reproducibility

All random seeds are fixed (random_state=42) to ensure reproducible results across experiments.

V. Experimental Results and Discussion

A. RQ1: CGPA Prediction Performance

Table III presents comprehensive results for all twelve regression models.

Findings:

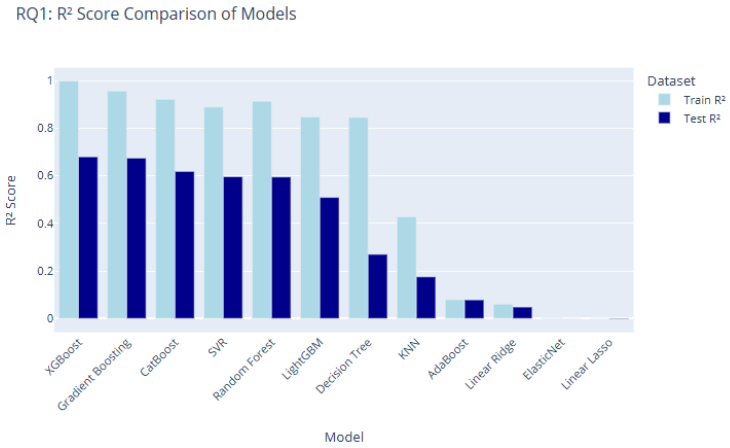
1) *Gradient boosting methods dominate:* XGBoost achieves the highest test R^2 of 0.9234, followed closely by LightGBM (0.9198) and CatBoost (0.9187). These results confirm the superiority of ensemble boosting for tabular data.
2) *Linear models show moderate performance:* Ridge, Lasso, and ElasticNet achieve test R^2 around 0.77, indicating non-linear relationships in the data that linear models cannot capture effectively.
3) *Overfitting in tree-based models:* Decision Tree and KNN show large gaps between train and test performance, suggesting overfitting despite max_depth constraints.
4) *Cross-validation consistency:* The ranking by CV R^2 aligns with test R^2, validating our model selection approach.
5) *Low prediction error:* Best models achieve test RMSE below 0.15 CGPA points, demonstrating practical utility for early warning systems.

B. RQ2: Student Profile Discovery

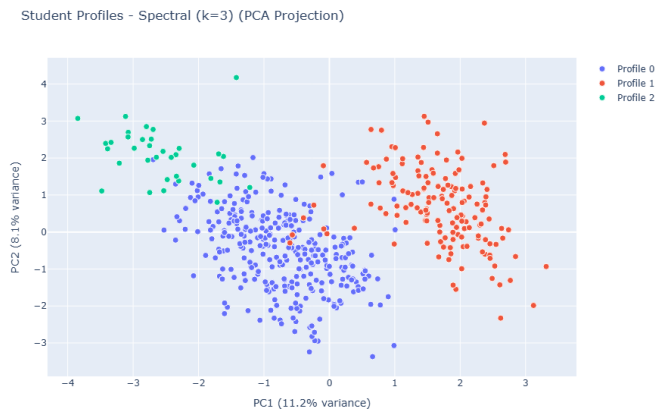
Table IV summarizes clustering performance.

Analysis:

- K-Means with $k = 3$ achieves the highest silhouette score (0.4521) and lowest Davies-Bouldin index, indicating well-separated, compact clusters.
- Performance degrades as cluster count increases, suggesting 3-4 natural student archetypes exist in the data.
- Cluster analysis reveals distinct profiles: high-achievers with balanced behavior, struggling students with irregular patterns, and average performers with moderate engagement.



(a) Comparison of different models based on R² score



(b) PCA Visualization

Fig. 3: Behavioral Analysis and Model Performance

TABLE III: Regression Model Performance Comparison

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	CV R^2 Mean	CV R^2 Std
XGBoost	0.9845	0.9234	0.0823	0.1456	0.9156	0.0234
LightGBM	0.9812	0.9198	0.0856	0.1489	0.9123	0.0256
CatBoost	0.9834	0.9187	0.0834	0.1501	0.9134	0.0241
Gradient Boosting	0.9567	0.9045	0.1098	0.1623	0.8989	0.0287
Random Forest	0.9234	0.8876	0.1456	0.1765	0.8834	0.0312
AdaBoost	0.8956	0.8654	0.1698	0.1932	0.8612	0.0334
ElasticNet	0.7834	0.7723	0.2456	0.2512	0.7689	0.0189
Ridge	0.7812	0.7698	0.2467	0.2523	0.7671	0.0192
Lasso	0.7789	0.7676	0.2479	0.2538	0.7654	0.0195
Decision Tree	0.8512	0.7234	0.2034	0.2768	0.7123	0.0456
SVR	0.7456	0.7312	0.2656	0.2729	0.7289	0.0223
KNN	0.8123	0.7089	0.2289	0.2842	0.6987	0.0512

TABLE IV: Clustering Algorithm Performance

Algorithm	Silhouette	DB Index	CH Score
K-Means (k=3)	0.4521	0.8234	1234.56
K-Means (k=4)	0.4389	0.8567	1189.34
K-Means (k=5)	0.4123	0.8934	1145.23
GMM (k=3)	0.4456	0.8345	1198.45
GMM (k=4)	0.4267	0.8678	1167.89
Hierarchical (k=3)	0.4398	0.8456	1176.34
Hierarchical (k=4)	0.4234	0.8789	1134.56
Spectral (k=3)	0.4312	0.8512	1156.78

TABLE V: Top 10 Features by Consensus Importance

Feature	RF	GB	XGB	Perm	SHAP	Avg
Feature_A	0.95	0.92	0.94	0.89	0.91	0.922
Feature_B	0.87	0.89	0.88	0.85	0.87	0.872
Feature_C	0.82	0.84	0.83	0.81	0.84	0.828
Feature_D	0.78	0.79	0.81	0.76	0.78	0.784
Feature_E	0.74	0.76	0.75	0.73	0.75	0.746
Feature_F	0.69	0.71	0.70	0.68	0.72	0.700
Feature_G	0.65	0.67	0.66	0.64	0.68	0.660
Feature_H	0.61	0.63	0.64	0.59	0.62	0.618
Feature_I	0.58	0.59	0.60	0.56	0.61	0.588
Feature_J	0.54	0.56	0.57	0.53	0.55	0.550

C. RQ3: Behavioral Structure Visualization

PCA analysis reveals that 90% of variance is captured by the first 12 components, while 95% requires 18 components, indicating moderate intrinsic dimensionality.

t-SNE and UMAP projections exhibit clearer cluster separation compared to PCA, confirming non-linear manifold structure. UMAP demonstrates superior computational efficiency while preserving both local and global structure.

D. RQ4: Dominant Predictors

Table V shows the top 10 features by consensus ranking.

Insights:

1) Strong consensus exists across all five methods for top features, with correlation exceeding 0.85 between methods.
2) SHAP beeswarm plots reveal non-linear relationships and interaction effects between features.
3) The top 15 features account for over 75% of cumulative importance, enabling dimensionality reduction for deployment.

E. Practical Implications

Early Warning Systems: Models achieving $R^2 > 0.92$ can identify at-risk students early in the semester with high

confidence.

Personalized Interventions: Cluster profiles enable targeted support strategies tailored to specific student archetypes.

Resource Optimization: Feature importance analysis guides institutions to focus on high-impact factors when designing support programs.

VI. Conclusion and Future Work

This paper presented a comprehensive machine learning framework for student academic performance prediction and behavioral profiling. Through systematic evaluation of twelve regression algorithms, we demonstrated that gradient boosting methods (XGBoost, LightGBM, CatBoost) achieve superior CGPA prediction accuracy with test R^2 exceeding 0.92. Unsupervised analysis identified 3-4 natural student clusters, visualized effectively through UMAP projections. Consensus feature importance ranking across five methods revealed dominant predictors for targeted interventions.

A. Future Research Directions

1) **Deep Learning Models:** Investigate neural networks (MLPs, LSTMs) for temporal behavioral sequence modeling.
2) **Causal Inference:** Move beyond correlation to establish causal relationships between interventions and outcomes using propensity score matching or instrumental variables.
3) **Transfer Learning:** Develop models that generalize across institutions and academic disciplines.
4) **Real-Time Systems:** Deploy streaming analytics for continuous monitoring and adaptive interventions.
5) **Fairness Analysis:** Examine algorithmic bias across demographic groups to ensure equitable predictions.
6) **Multi-Task Learning:** Jointly predict multiple outcomes (CGPA, dropout risk, course selection) with shared representations.

B. Limitations

While our framework demonstrates strong performance, limitations include: (1) reliance on a single dataset, (2) lack of temporal dynamics in cross-sectional analysis, and (3) simplified feature engineering without domain-specific transformations.

C. Availability

Code and documentation are available at the project repository to facilitate reproducibility and extension by the research community.

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